Embedding comparison of BERT and ChatGPT answers of «General Knowledge» dataset

General Knowledge dataset

```
Total (non alpaca): 6315
- Facts - 80.8 %
    - Nature - 16.5 %
    - AI, Computer science, Robotics - 7.3 %
    - Physics, Chemistry - 16.3 %
    - Geography, History - 11.2 %
    - People - 16 %
    - Sports - 13.5 %
- Recommendation, Reasoning, Dilemma - 17.8 %
- Others - 1.4 %
```

- A set of ~38000 questions and answers.
- Is meant to be used for model training.
- e.g.: {
  'Question': 'What is the largest species of shark',
  'Answer': 'The whale shark is considered the largest species of shark, with adults reaching lengths of up to 40 feet or more and weighing several tons.'
  }
- https://huggingface.co/datasets/MuskumPillerum/General-Knowledge

E5 small v2 model

- Text Embeddings model
- Model has 12 layers.

https://huggingface.co/intfloat/e5-small-v2

Input texts have to be labeled with query and answer:

```
input_texts = ['query: how much protein should a female eat',
               'query: summit define',
               "passage: As a general guideline, the CDC's average requirement of
               "passage: Definition of summit for English Language Learners. : 1
```

GPT API Querys

- Querys have been done with «gpt-4o» model.
- Threading & async.
- Delay had to be implemented for «Rate limit exceeded».

GPT API Querys

- First try: $12 used within 20min then connection lost. No data saved.
- Second try: much slower queries, data streamed to csv continuously. Script interrupted after hours and only 743 answers received.
- Total of $32 spent.

Dataset merged

```
1  df
```

✓ 0.0s Open 'df' in Data Wrangler

	Answer	Question	Response
0	Artificial Intelligence refers to the developm...	What is Artificial Intelligence?	Artificial Intelligence (AI) refers to the sim...
1	The two main categories of Artificial Intellig...	What are the two main categories of Artificial...	The two main categories of Artificial Intellig...
2	Machine Learning is a subset of Artificial Int...	What is Machine Learning?	Machine learning is a subset of artificial int...
3	Deep Learning is a subset of Machine Learning ...	What is Deep Learning?	Deep learning is a subfield of machine learnin...
4	Natural Language Processing is a subset of Art...	What is Natural Language Processing?	Natural Language Processing (NLP) is a subfiel...
...	...	...	...
5996	The five medicinal plants widely used in Ayurv...	Name five medicinal plants widely used in Ayur...	Five medicinal plants widely used in Ayurveda ...
5997	A simple solution is to traverse the binary tr...	generate an algorithm to find the first common...	To find the first common ancestor of two nodes...
6004	Verse 1\nC G F C\nB-flat F G C\n\nChorus...	Compose a song in C Major.?	Here's a simple song in C Major for you. It in...
6007	-625 is a negative integer.	What type of number is -625??	The number \(-625\) is an integer, as it is a ...
6010	One chemical that is both an acid and a base i...	Name one chemical that is both an acid and a b...	One chemical that acts as both an acid and a b...

743 rows × 3 columns

The responses from GPT4o have been merged back with the dataset and incompletes have been dropped. Then saved as parquet.

Embedding

```python
# Average-Pooling-Funktion von der Huggingface-Quelle
def average_pool(last_hidden_states: Tensor,
                 attention_mask: Tensor) -> Tensor:
    last_hidden = last_hidden_states.masked_fill(~attention_mask[..., None].bool(), 0.0)
    return last_hidden.sum(dim=1) / attention_mask.sum(dim=1)[..., None]

# Tokenizer und model von der Huggingface-Quelle
tokenizer = AutoTokenizer.from_pretrained('intfloat/e5-small-v2')
model = AutoModel.from_pretrained('intfloat/e5-small-v2')

# Vorbereiten des tokenizers für pooling
tokenizer_kwargs = {
    'max_length': 80,
    'padding': 'max_length',
    'truncation': True,
    'return_tensors': 'pt'
}
tokenizer_with_args = partial(tokenizer, **tokenizer_kwargs)

# Tokenize the input texts
from itertools import repeat
from concurrent.futures import ProcessPoolExecutor
import torch
device = torch.device("cuda" if torch.cuda.is_available() else "cpu")
model.to(device)

# ProcessPoolExecutor zum umgehen des Global Interpreter Lock
def pooling(section):
    results = []
    with ProcessPoolExecutor() as executor:
        for i, tokens in enumerate(executor.map(
            tokenizer_with_args, section
            )):
            results.append(tokens)
    # Tensoren müssen wieder kombiniert werden mit torch.cat
    combined_batch_dict = {
        key: torch.cat([torch.tensor(batch[key]) for batch in results], dim=0)
        for key in results[0]
    }

    return combined_batch_dict

batch_dict = pooling(input_texts)
```

- Average pooling as suggested by huggingface.
- Cuda & Multiprocessing was necessary to avoid out-of-memory error.
- torch.cat() for concatenation of the tensors.

Scores & Ranking

- Cosine similarities of the word embedding vectors of the questions and answers are collected in sorted lists.
- Correct answer for each question should be on top.
- Metric: true rank of the correct answer in the list.
- Smaller rank is better.

Scores & Ranking

- What is the average rank of the correct answer?
- Cosine similarity with:
  torch.matmul(Questions, Answers.T)
- Mean rank (low): sum(ranks) / len(ranks)
- Mean reciprocal rank (high):
  sum(1.0 / rank for rank in ranks) / len(ranks)
- Precision@1 (high):
  sum(1 for rank in ranks if rank <= k) / len(ranks)

```
Mean Rank: 4.658142664872l4
MRR: 0.7916729202811885
Precision@1: 0.6796769851951547
GPT Mean Rank: 2.3485868102288023
GPT MRR: 0.7867026154182879
GPT Precision@1: 0.6729475100942126
```

Lessons learned

- Need to investigate, why GPT Api usage failed so badly.
- Concurrency methods helped a lot, but also wasted a lot of my budget in this case.